



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Dataset Overview

3,900

Total Purchases

Transactions analyzed

18

Data Points

Features per transaction

37

Missing Values

In review ratings only

Customer Demographics

- Age, Gender, Location
- Subscription Status

Shopping Behavior

- Purchase patterns & frequency
- Discounts & promotions
- Review ratings & shipping

Data Preparation Process

01

Data Loading & Exploration

Imported dataset using pandas, analyzed structure with `df.info()` and summary statistics

02

Missing Data Handling

Imputed Review Rating nulls using median rating per product category

03

Column Standardization

Renamed columns to snake_case for consistency and readability

04

Feature Engineering

Created age_group bins and purchase_frequency_days columns

05

Data Consistency Check

Verified redundancy, dropped promo_code_used column

06

Database Integration

Connected to PostgreSQL and loaded cleaned data for SQL analysis

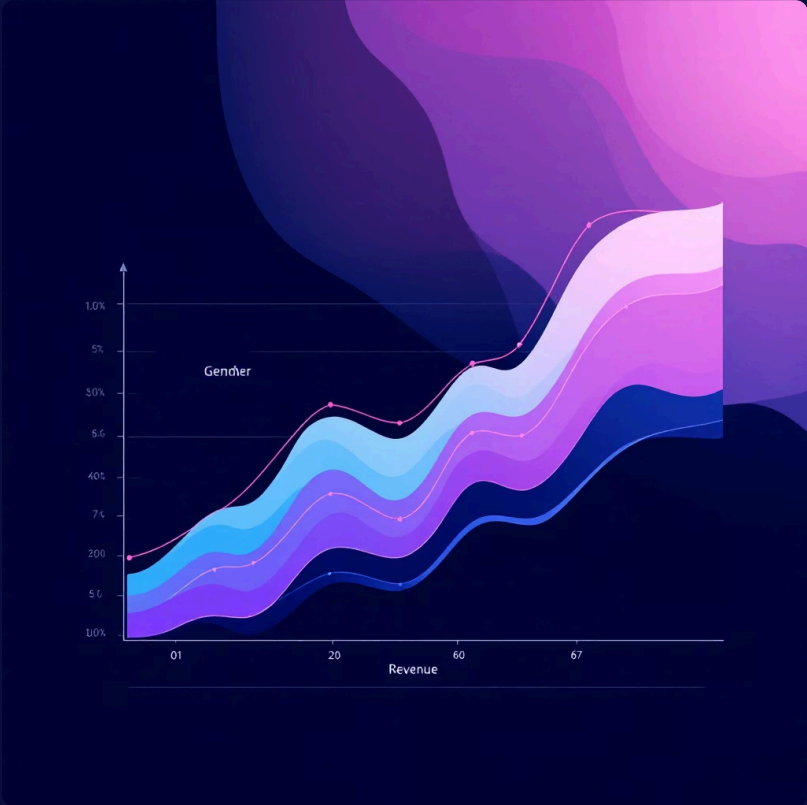
Revenue Insights

Revenue by Gender

Male customers generate slightly higher total revenue compared to female customers

Subscribers vs. Non-Subscribers

Subscribers show higher average spend and contribute significantly more to total revenue



Express Shipping

Higher average purchase amounts than Standard shipping

Discount Users

Many discount users still spend above average purchase amount

Top-Rated Products

Products with highest average review ratings drive customer satisfaction

1

Jewelry

Highest customer satisfaction scores across all categories

2

Blouse

Strong ratings indicate quality and fit excellence

3

Dress

Consistent positive feedback from customers

4

Sweater

High marks for comfort and style

5

Pants

Reliable quality drives repeat purchases

Discount-Dependent Products

Five products with highest percentage of discounted purchases



Coat

Heavily reliant on promotions

Jacket

Strong discount dependency

Boots

Frequent promotional sales

Handbag

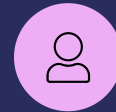
Discount-driven purchases

Sunglasses

Price-sensitive category

📌 **Key Insight:** These products may need pricing strategy review to balance sales volume with profit margins

Customer Segmentation



New Customers

First-time buyers exploring product range

- 1-2 purchases
- Building trust



Returning Customers

Engaged shoppers with repeat behavior

- 3-5 purchases
- Growing loyalty

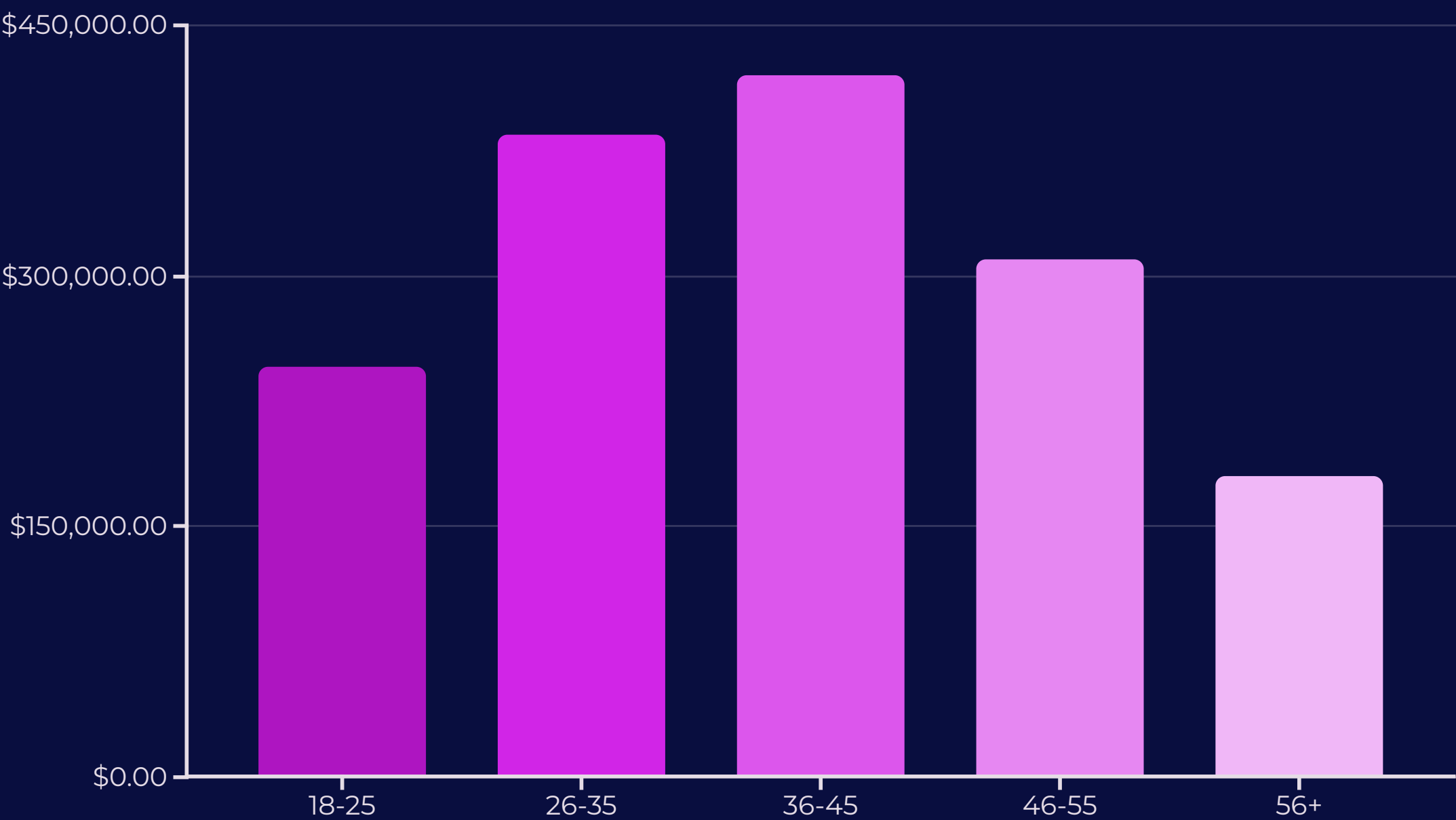


Loyal Customers

Brand advocates with consistent purchases

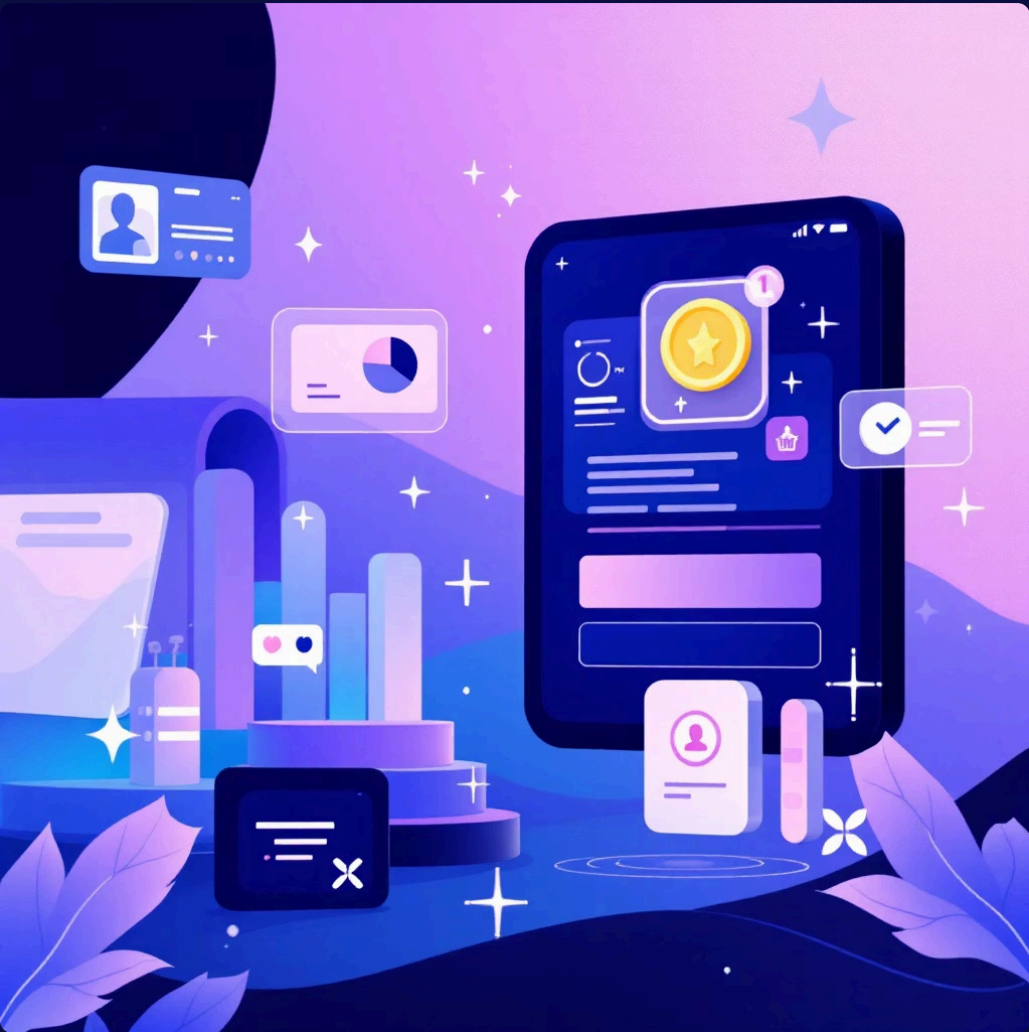
- 5+ purchases
- High lifetime value

Revenue by Age Group



Ages 36-45 represent the highest revenue-generating segment, followed closely by 26-35 age group

Subscription & Loyalty Connection

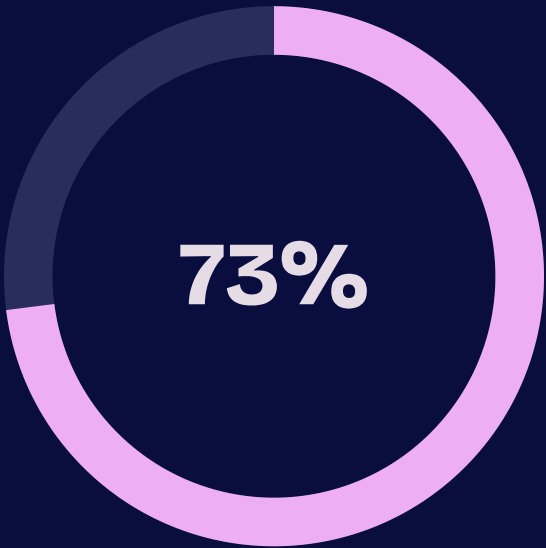


Key Finding

Customers with **5+ purchases** show significantly higher subscription rates

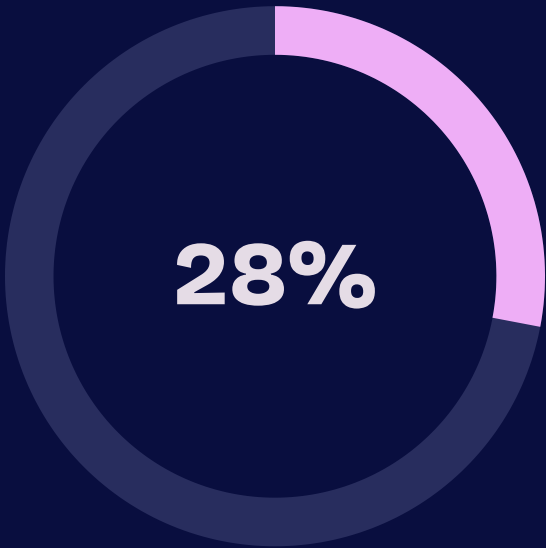
Repeat Buyers

- More likely to subscribe
- Higher average order value
- Stronger brand affinity



Subscription Rate

Among customers with 5+ purchases



Subscription Rate

Among customers with fewer purchases

Strategic Recommendations

1

Boost Subscriptions

Promote exclusive benefits and perks for subscribers to increase conversion

2

Customer Loyalty Programs

Reward repeat buyers to accelerate movement into "Loyal" segment

3

Review Discount Policy

Balance promotional sales boosts with margin control for sustainable growth

4

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns

5

Targeted Marketing

Focus efforts on high-revenue age groups (36-45) and express-shipping users